Document Identifier: **PMC8331171**

SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
Field Sample Permit: After 10-20 d of water deprivation , the phenotype was observed , and a sample collection or re-watering treatment for 2-3 d was performed when needed.
Inclusion and Exclusion Criteria
not required.
Attrition
not required.
Sex as a biological variable
not detected.
Subject Demographics
Age: Drought treatment For each independent biological experiment , 12-15 4-week-old cotton seedlings (having at least two fully expanded true leaves) were subjected to a cessation in watering .
Randomization
not detected.
Blinding
not detected.
Power Analysis
not detected.
Replication
The VIGS assay for GhDAN1 was performed with at least five biological replicates .
Type: The VIGS assay for GhDAN1 was performed with at least five biological replicates .
Number: The VIGS assay for GhDAN1 was performed with at least five biological replicates .

Data Information	
Availability: Accession numbers Novel data generated in this study, including raw data of mRNA-seq and ChIRP-seq, have been deposited in the National Center for Biotechnology Information under the accession number PRJNA637977.	
Identifiers: Accession numbers Novel data generated in this study, including raw data of mRNA-seq and ChIRP-seq, have been deposited in the National Center for Biotechnology Information under the accession number PRJNA637977.	PRJNA637977
Identifiers: Pol II ChIP-seq data are under the accession number PRJNA373801, as previously reported (Zhao et al., 2018a, 2018b).	PRJNA373801

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
Anti-histone H3 (trimethyl K4) antibodies (ab8580) and anti-histone H3 (trimethyl K27) antibodies (ab6002) were used in the assay.	Anti-histone H3 (trimethyl K4)		
	anti-histone H3		
Recombinant DNA			
Fragments of candidate genes were amplified and separately cloned into a pTRV2 vector by the infusion method to construct recombinant pTRV2 vectors .	pTRV2		Suggestion: RRID:Addgene_105349(link)
Of these , the recombinant pTRV2-GhCLA vector was used as a positive control in order to monitor the effect of the VIGS.	pTRV2-GhCLA		
The vectors , including pTRV1 , recombinant pTRV2 , and pTRV2 empty vectors , were then separately introduced into specimens of A . tumefaciens strain GV3101 by electroporation .	pTRV1		
A GhDAN1 fragment from +9 to +186 bp was first cloned into the pMD19-T vector to construct pMD19-T-GhDAN1 for sequence confirmation .	pMD19-T		
	pMD19-T-GhDAN1		
Software and Algorithms			
The remaining transcripts were subjected to HMMER version 3.1b2 to exclude the transcripts containing known protein domains (cutoff <0.001 Finn et al. , 2011) .	HMMER		Suggestion: Hmmer (RRID:SCR_005305)(link)
Third , the peaks shared by the odd and even pools (designated as ChIRP valid peaks) were generated using intersectBed in the BEDTools suite (Quinlan and Hall , 2010) .	BEDTools		Suggestion: BEDTools, RRID:SCR_006646(link)
Two biological replicates for ChIRP reaction were performed , and similar AAAG rich motif was repetitively established by MEME software.	MEME		Suggestion: MEME Suite - Motif-based sequence analysis tools (RRID:SCR_001783)(link)
Gene expression was calculated using StringTie with parameter (-A) (Pertea et al., 2016).	StringTie		Suggestion: StringTie (RRID:SCR_016323)(link)
For differential expression analysis, each gene count was calculated using HTseq-count (Anders et al., 2015).	HTseq-count		
DEGs were tested using the edgeR package (Robinson et al., 2010).	edgeR		

NGS reads were mapped to the cotton genome using Bowtie, allowing up to two mismatches (Langmead et al., 2009).	Bowtie		Suggestion: Bowtie (RRID:SCR_005476)(link)
For the uniquely mapped reads, peak calling of the ChIP-seq data was performed using MACS2 (Zhang et al., 2008).	ChIP-seq		
Phylogenetic analysis Phylogenetic tree depicting the evolutionary relationship of DAN1 in cotton lineage was performed using MEGA software (Tamura et al., 2007).	MEGA		
Statistical analyses All statistical analysis was done with the software Graphpad Prism (GraphPad Software, La Jolla, CA, USA).	Graphpad Prism		Suggestion: GraphPad Prism (RRID:SCR_002798)(link)
	GraphPad		Suggestion: GraphPad Prism, RRID:SCR_002798(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
The Students t test, Mann-Whitney U-test, or Fishers exact test was used whenever appropriate.	Students t test
	Mann-Whitney U-test
	Fishers exact test

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For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq